



JAVAD EBRAHIMIZADEH

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EXPERT AND INTERESTS

- Massive MIMO for Wireless Communication Systems (5G and Beyond)
- Millimeter-wave Channel Modelling and Characterization
- Hybrid Phased Array Architecture and Beamforming
- Microwave and Millimeter-wave Antenna
- Signal processing and machine learning

EDUCATION

Doctor of Philosophy <i>Wireless propagation & Active phased array system</i> KULeuven University	2021 – Present Leuven, Belgium
Master of Science <i>Microwave Fields & Waves</i> University of Tehran	2012 – 2015 Tehran, Iran
Bachelor of Science <i>Telecommunication</i> University of Sistan and Baluchestan	2008 – 2012 Zahedan, Iran

WORK EXPERIENCES

Massive MIMO Signal processing SONY research center	Jan 2024-March 2024 SONY, Lund, Sweden
• Massive MIMO signal processing using machine learning. • Internship at SONY company for SVD-Based Channel State Estimation Feedback Enhanced by Neural Network for Massive MIMO.	
On Glass Antenna AGC Glass Company	2020 – 2021 Charleroi, Belgium
• Design and simulate a wire antenna on the realistic backshield glass of the car for AM, FM, DAB, and TV bands using Altair FEKO. • Fabricate and tune the wire antenna on a realistic car in an anechoic chamber	
Microwave Sensor Angstrom Laboratory	2018 – 2019 Uppsala, Sweden
• Microwave imaging for bone density assessment (osteopenia). • Design and fabricate UWB on-body antenna for fat-body communication	
Applied Electromagnetic Research Center University of Tehran	2016 – 2018 Tehran, Iran
• Testing and certifying different types of antennas and RF active and passive c by using several advanced experimental instruments • Working with several advanced experimental instruments such as: PNA Microwave Network Analyzer, PSG Analog Signal Generator, Agilent spectrum analyzer, and Probing Station. Conducting various measurements such as 2-D and 3-D measurement of the antenna radiation pattern, peak gain, RCS, and frequency response of RF circuits	

SKILLS

Software:

HFSS, CST, Altair Feko, Altium designer, ADS, System vue, and COMSOL

Programming:

Python, MATLAB, and Mathematica

RF Instruments:

VNA, Oscilloscope, Spectrum Analyzer, Power meter, Anechoic Chamber, and USRP software-defined radio

NOTABLE PROJECTS

RCS Simulator Software (PO-GO method accelerated with machine learning) | *Python* 2024

SONY company and University of KULeuven, Belgium

- A Python-based software has been developed to facilitate the calculation of bistatic RCS scattering using the Physical Optics-Geometric Optics (PO-GO) accelerated with a feedforward neural network (FFNN).
- Developing a feedforward neural network for estimating channel state information.

Active phased array system | *MATLAB, HFSS, CST, ADS, System vue* 2023-2024

University of KULeuven, Belgium

- Design and fabricate a dual-beam phased array for Satellite Communications. A 64-element uniform rectangular array at X-band with the capability of generating dual concurrent and independent directive beams

Millimeter-wave MIMO channel measurement | *MIMO-mmWave Channel sounder* 2023

LUND University, Sweden

- Using ray tracing method, millimeter-wave channel modeling and characterization. The measurement campaign for simulation validation using a MIMO channel sounder at 28 GHz. Post-processing the measured data for multipath component classification

Wireless channel modeling and characterization | *Python, MATLAB* 2021

University of KULeuven, Belgium

- Developing a novel quasi-deterministic ray tracing (QD-RT) method for statistically analyzing the mmWave wireless channels. The QD-RT method benefits from low complexity and can generate the same statistical channel parameters as the deterministic ray tracing method

THESIS

Doctor of Philosophy | *KULeuven university* 2023

Title: Supervisor: Prof. Guy Vandenbosch

- Statistical channel model for street canyon scenario (simulation and measurement)
- To design a low complexity mmWave multi-beam phased array antenna

Master of Science | *University of Tehran* 2015

Title: Supervisor: Prof. Mojtaba Dehmollaian

- Electromagnetic time-reversal imaging of pinholes in pipes

Bachelor of Science | *University of Sistan and Baliuchestan* 2015

Title: Supervisor: Prof. Mehdi Rezaei

- Wireless Communication Using ARM Microcontroller

MAIN PUBLICATIONS

Statistical 3D millimeter-wave channel modeling IEEE Transactions on Antennas and Propagation	2023 Published
RCS-based quasi-deterministic ray tracing for statistical channel modeling The 26th European Microwave Week (EuMW 2023)	2023 Published
Polarimetric bistatic RCS distribution: A sensing metric for 5G wireless communication IEEE globecom workshops (GC Wkshps)	2023 To be Published
low-loss multi-beam large phased array architecture International Symposium on Antennas and Propagation	2023 Published
Millimeter-wave Scattering from Building Facade: A Simulation and Verification Study European Conference on Antennas and Propagation	2024 To be Published
Accelerating PO-GO Method by Machine Learning for Calculating RCS EuMW 2024	2024 Submitted
SVD-Based CSI Feedback Enhanced by Neural Network for Massive MIMO Signal Processing	2024 Tobe Submitted
Electromagnetic time-reversal imaging of pinholes in pipes IEEE Transactions on Antennas and Propagation	2016 Published

CONFERENCES AND PRESENTATIONS

Neural network for massive MIMO application Internship, SONY, Sweden	2024
18th European Conference on Antennas and Propagation (EuCAP) Conference Presentation, Glasgow, Scotland	2024
The 26th edition of the European Microwave Week Conference presentation, Berlin, Germany	2023
International Symposium on Antennas and Propagation (AP-S/URSI) Conference Presentation Portland, Oregon, USA	2023
Practical Aspects of Private Wireless Networks at Nokia Bell Lab Seminar, Espoo, Finland	2023
Summer School of Information Engineering Summer school, Rome, Italy	2022
CoNEXT 2022 Conference attending, Rome, Italy	2022
3GPP standard training Seminar, Rome, Italy	2023
13th European Conference on Antennas and Propagation (EuCAP) Conference Presentation, Krakow, Poland	2019

TEACHING EXPERIENCES

Electromagnetic compatibility (EMC) <i>KULeuven university, Belgium</i>	2023
Robotic lab teaching assistance <i>Uppsala university</i>	2018 – 2019

Antenna lab teaching assistance <i>Uppsala university</i>	2018 – 2019
Advanced mathematic <i>University of Tehran</i>	2014 – 2015

HONORS AND AWARDS

Marie Sklodowska-Curie Scholarship	2021
MINTS - European Training Network on Millimeter-wave NeTworking and Sensing for Beyond 5G	
Iranian patent	2015
Best inventor and thesis among master students of the University of Tehran	
University entrance exam	2012
Ranked 47th among 75,000 participants in the Iranian universities entrance exam nationwide	

COMMUNITY INVOLVEMENT

Whattrips hiking memebr	2022 – 2023
Hike and explore the breathtaking natural landscapes of Europe	Leuven, Belgium